

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**A COGNITIVE TWO-TIER ARCHITECTURE FOR
AUTOMATED THREAT DETECTION AND
CONTEXTUAL RESPONSE USING AGENTIC AI**

by

Nguyen Duc Binh

Student ID: 24MSE13183 – Class: MSE23HN

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Supervisor:

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Dr. Dang Van Hieu

Abstract

In the era of digital transformation and burgeoning network traffic, modern Security Operations Centers (SOCs) face severe alert fatigue, as the volume of telemetry collected from monitoring systems far exceeds manual analysis capacity. Traditional defense tools exhibit clear limitations: static signature rules generate high false-positive rates due to a lack of context interpretation, while pre-programmed SOAR playbooks execute reliably only on anticipated scenarios. Transitioning to Large Language Models (LLMs) enables automated contextual reasoning in natural language; however, operational adoption remains constrained by latency bottlenecks, high computational costs, general adversarial risks (typically log-substrate prompt injections), and AI hallucinations. To address these challenges, this thesis proposes SENTINEL, a cognitive two-tier security architecture operating fully air-gapped, designed on the principle of placing cheap decisions before expensive ones. Tier 1 combines a Welford $\mathcal{O}(1)$ statistical filter and a LightGBM Machine Learning Gate to settle stream noise at wire-speed, offloading 97.5% of telemetry without invoking LLM resources and reducing median end-to-end latency from 17,174.7 ms to 0.88 ms. Tier 2 executes a LangGraph agentic engine integrated with a Dual-RAG hybrid retrieval mechanism (FAISS + BM25 RRF $k = 60$) over 433 MITRE ATT&CK technique entries, while the AWAIT_HITL deferral channel reduces analyst workload by 84.24% while retaining a 95.0% recall on genuine threats. Finally, Delimited Data Encapsulation provides structural guarantees to robustly withstand 678 adversarial attack samples (including log-substrate prompt injections), and HMAC-SHA256 signature chaining ensures effective tamper-detection for forensic audit trails.

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List of Abbreviations

Abbreviation	Full Description
CTI	Cyber Threat Intelligence
ECDF	Empirical Cumulative Distribution Function
FAISS	Facebook AI Similarity Search
GGUF	GPT-Generated Unified Format
HITL	Human-in-the-Loop
HMAC	Hash-based Message Authentication Code
IDS / IPS	Intrusion Detection / Prevention System
LLM	Large Language Model
MITRE ATT&CK	MITRE Adversarial Tactics, Techniques, and Common Knowledge
RAG	Retrieval-Augmented Generation
RAGAS	Retrieval-Augmented Generation Assessment
RRF	Reciprocal Rank Fusion
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation, and Response
SOC	Security Operations Center
WAF	Web Application Firewall

Chapter 1

Introduction

This chapter states the research context and central problem of SENTINEL, a cognitive two-tier architecture automating threat detection and incident response in Security Operations Centers (SOCs), together with the objectives, research questions, research object, scope, and scientific contributions of the thesis.

1.1 Background and Motivation

In the era of digital transformation and burgeoning network traffic, modern Security Operations Centers (SOCs) face severe alert fatigue, as daily telemetry ingestion reaching millions of log events far exceeds manual analysis capacity [1]. To avoid missing intrusions, monitoring sensors are configured toward high sensitivity, causing the vast majority of queued alerts to represent benign activity that does not require direct analyst intervention [2, 3]. Consequently, the operational bottleneck of a modern SOC lies not in data ingestion, but in allocating the scarce attention of human analysts.

Three traditional defence layers—firewalls with static signature intrusion detection (IDS/IPS), SIEM platforms correlating telemetry via rule sets, and pre-programmed SOAR playbooks—exhibit clear limitations: they lack dynamic contextual reasoning and execute reliably only on anticipated scenarios [4]. Unresolved events accumulate in manual queues, forcing analysts to perform time-consuming tasks: extracting raw logs, reconstructing source IP behaviors, cross-referencing asset inventories, and mapping attack techniques. This sequential process creates both a time bottleneck causing backlog accumulation and a context bottleneck as deciding evidence remains scattered across unstructured log records.

Transitioning to Large Language Models (LLMs) enables automated contextual reasoning in natural language; however, operational adoption is constrained by three major challenges: (1) multi-second LLM inference latency conflicts with wire-speed data streams [5]; (2) the AI reasoning layer itself becomes vulnerable to log-substrate prompt injection attacks [6], mandating tamper-sealed audit trails; and (3) internal security telemetry must execute fully local (air-gapped) on finite hardware due to strict data privacy requirements [7]. The 4-to-5 order-of-magnitude computational cost gap between traditional statistical rules ($\ll 1$ ms) and LLMs (seconds) necessitates designing SENTINEL on the principle of *placing cheap decisions before expensive ones*: offloading stream noise at wire-speed and invoking LLM reasoning only for events that truly warrant complex analysis.

1.2 Objectives, Research Questions, Scope and Contributions

To address the central problem, the general objective of this thesis is structured into three distinct yet mutually supportive research tracks, mapped to three research questions (RQs) and primary scientific contributions. The complete architectural design and quantitative experimental evaluations are demonstrated in Chapter 4.

The first track focuses on telemetry tiering: designing a stateless dual-stage filtration pipeline at wire-speed to offload telemetry at the cheap computing tier. RQ1 is formulated as follows: *How can a multi-tiered architecture be designed to autonomously offload security telemetry at wire-speed, thereby addressing the latency bottlenecks and resource consumption inherent in Large Language Models?* The corresponding contribution quantifies the principle of placing cheap decisions before expensive ones and reports the offload rate against the attack base rate at which it was measured.

The second track focuses on reasoning security and audit integrity: protecting the language model via structural guardrails and sealing forensic records. RQ2 is formulated as follows: *What methodologies and security mechanisms can effectively withstand adversarial risks to protect the AI's reasoning flow, while ensuring the integrity and tamper-evidence of forensic audit trails?* The corresponding contribution introduces Delimited Data Encapsulation,

providing deterministic structural guarantees independent of probabilistic classifiers [8].

The third track focuses on cognitive reasoning and incident triage: building a stateful agentic engine integrated with domain security knowledge. RQ3 is formulated as follows: *How can stateful reasoning capabilities and domain-specific knowledge retrieval be integrated into an AI Agent to automate intrusion analysis, attribute attack techniques, and generate transparent forensic reports?* The corresponding contribution provides a stateful LangGraph agent operating over a STIX-aligned MITRE ATT&CK knowledge base, coupled with a RAG-anchoring guardrail to mitigate hallucinations.

The scope and methodology are bounded as follows: the research domain focuses on automated telemetry analysis, alert triage, and AI reasoning security in SOCs. The research object encompasses high-speed telemetry tiering algorithms (application signature Rule-based, Welford $\mathcal{O}(1)$ statistics coupled with LightGBM Machine Learning Gate), deterministic adversarial defense (Delimited Data Encapsulation and HMAC-SHA256), and stateful reasoning agents (LangGraph FSM) with hybrid retrieval (Dual-RAG FAISS + BM25 RRF). Experimentally, SENTINEL is evaluated on three benchmark datasets, CSE-CIC-IDS2018 [9], CSIC 2010 [10] and DAPT 2020 [11], alongside 823 adversarial attack samples (AdvBench [12], deepset [13], jackhhao [14]) deployed on an air-gapped workstation.

1.3 Structure of the Thesis

Chapter 2 systematizes theoretical foundations and surveys related work to establish the research gap. Chapter 3 presents the research methodology and technical design of SENTINEL. Chapter 4 presents the quantitative evaluation in three sections, one for each research question, in the order stated above. Chapter 5 concludes with a summary of contributions, discussion of limitations, and directions for future work.

Chapter 2

Theoretical Background and Literature Review

This chapter presents the foundational theoretical principles of this thesis, covering: current SOC operations and the cognitive bottleneck, deterministic rules, online statistics, and Tier 1 machine learning, local large language models and Tier 2 agentic intelligence, adversarial AI security and log audit cryptography, and an overview of related work to delineate the research gap addressed by SENTINEL.

2.1 Current SOC Operations and the Cognitive Bottleneck

This section analyzes the core architectural causes of the alert overload described in Chapter 1. In a standard SOC model, monitoring relies on aggregating raw telemetry from sensors such as Next-Generation Firewalls (NGFW), Intrusion Detection Systems (IDS/IPS), and Endpoint Detection and Response (EDR) into a centralized SIEM platform [1], at volumes far exceeding manual human analysis capacity [15]. The primary architectural bottleneck stems from the operational nature of these sensors: they render decisions predominantly via static signatures or deterministic rule sets. Because the cost of missing an intrusion is severe, operators are forced to lower detection thresholds. However, lowering thresholds in signature-based systems exponentially increases false positive volume in analysis queues without enhancing contextual understanding [2, 3].

At the higher operational layer, Security Orchestration, Automation, and Response (SOAR) platforms attempt to reduce manual workload by executing pre-programmed response playbooks [4]. Nevertheless, SOAR is strictly constrained by playbook rigidity: a playbook triggers successfully only when alert parameters precisely match anticipated scenarios.

For complex multi-stage attack chains, novel behavioral anomalies, or unstructured evidence scattered across multiple records, SIEM, IDS/IPS, and SOAR layers collectively fail to form contextual judgments. Consequently, all unresolved cases escalate to manual analysis queues, transforming human analysts into the sole cognitive bottleneck of the entire security lifecycle.

2.2 Deterministic Rules, Online Statistics, and Tier 1 Machine Learning

Operating at wire-speed imposes strict algorithmic constraints on Tier 1: processing mechanisms must achieve constant $\mathcal{O}(1)$ time and memory complexity. First, a rule-based defense layer uses static set-matching and service-port lookups to instantly eliminate coarse anomalies without consuming computational resources. Next, baseline tracking applies Welford’s incremental recurrence formula [16] to continuously update the running mean μ_k and sum of squared deviations S_k per observation, computing standard deviation $\sigma_k = \sqrt{S_k/(k-1)}$ and dynamic Z-scores without storing raw historical telemetry:

$$\mu_k = \mu_{k-1} + \frac{x_k - \mu_{k-1}}{k}, \quad S_k = S_{k-1} + (x_k - \mu_{k-1})(x_k - \mu_k) \quad (2.1)$$

Finally, Gradient Boosted Decision Tree (GBDT) machine learning models—exemplified by LightGBM [17]—provide rapid classification of multi-dimensional telemetry patterns. Combining deterministic rule filtering, Z-score statistical normalization, and machine learning classification creates a high-speed multi-stage defense line, ensuring high-throughput offloading while maintaining robustness against adversarial telemetry variants.

2.3 Local Large Language Models and Tier 2 Agentic Intelligence

Executing the reasoning tier entirely on-premise is mandatory to guarantee internal security data sovereignty, while imposing strict GPU memory boundaries. Quantization

compresses weight representations onto a coarser integer grid, trading a minor amount of numerical fidelity for significant memory reduction [18]. Under GGUF Q4_K_M, the security-specialized Foundation-Sec-8B-Instruct model [19, 20] occupies approximately 7–8 GB VRAM, preserving sufficient headroom for context buffers and retrieval indices on a local GPU.

The reasoning tier orchestrates analysis flows via a LangGraph state graph [21]. To mitigate hallucinations and ground decisions in verifiable evidence [22], the system combines hybrid retrieval across dense vector search (FAISS) [23] and sparse keyword matching (BM25) [24]. Retrieved candidates are fused by rank position using Reciprocal Rank Fusion (RRF) [25]:

$$RRF_Score(d) = \sum_{m \in M} \frac{1}{k + r_m(d)} \quad (2.2)$$

enabling transparent cross-referencing of log telemetry against MITRE ATT&CK technique codes [26] and NIST SP 800-61r2 incident handling guidelines [27].

2.4 Adversarial AI Security and Log Audit Cryptography

Integrating Large Language Models into security operations introduces log-substrate prompt injection risks [6, 8]. Because raw log content is adversary-controlled, content-matching filters or probabilistic keyword detectors remain vulnerable to novel adversarial variants [28]. This theoretical foundation necessitates protecting the reasoning tier via deterministic structural isolation (Delimited Data Encapsulation), establishing an absolute boundary between control instructions and untrusted telemetry.

Regarding forensic audit integrity, evidence records are sealed in sequence using HMAC-SHA256 signature chaining [29] under a local secret key. Theoretical boundaries must be precisely stated: symmetric HMAC encryption guarantees tamper-evidence and modification detection, but does not provide non-repudiation, which requires asymmetric digital signature infrastructure.

2.5 Overview of Related Research and Research Gaps

AI-driven security operations research has evolved across three main architectural paradigms, as surveyed by Zhang et al. [30].

The first group, traditional enterprise defenses, comprises SIEM platforms such as IBM QRadar and SOAR tools like Palo Alto Cortex XSOAR [1, 4]. These systems are complemented by online log parsers like Drain [31], which compress syntax but cannot analyze behavioral context. Operators bear the burden of this rigidity: in interviews with SOC personnel, AlAhmadi et al. [3] identified poor alert quality as the primary driver of fatigue. The core strength of this baseline is low latency, a property SENTINEL’s Tier 1 inherits and refines.

The second group, retrieval-augmented classification agents such as CyberRAG [32], orchestrates domain-tuned classifiers behind an LLM agent, reporting over 94% per-class accuracy on web attacks. While effective at analyzing selected events, this architecture does not address front-end triage, lacking a wire-speed filtering layer to manage incoming event volume.

The third group, multi-agent and governance-focused frameworks, includes AutoBnB-RAG [33], LanG [34], and Policy-Guided Threat Hunting [35]. AutoBnB-RAG evaluates multi-agent incident response in simulated tabletop environments (*Backdoors & Breaches*). LanG utilizes LangGraph with human-in-the-loop checkpoints and dual-layer defense (regex plus Llama Prompt Guard 2 achieving 98.1% F1). Policy-Guided Threat Hunting employs reconstruction autoencoders and deep reinforcement learning prior to LLM contextual analysis.

A precise research gap emerges: cost-tiering exists in Policy-Guided Threat Hunting, prompt defense exists in LanG, and local execution is demonstrated across systems. However, no single architecture unifies: (1) resolving the vast majority of telemetry at a statistical/ML tier with sub-millisecond latency and offload rates reported against the attack base rate at which they were measured; (2) stateful reasoning on a local GPU under strict

retrieval grounding rules; and (3) structural prompt defense paired with HMAC-SHA256 forensic sealing. Table 1 provides a detailed comparative analysis.

Table 1. Comparative analysis between SENTINEL and existing security defense solutions

Evaluation Criteria	SIEM / SOAR (QRadar, XSOAR)	LLM Agents (CyberRAG, LanG, Policy-Guided)	SENTINEL Architecture (Thesis)
Core Technique	Static correlation rules, rigid Python/JS scripts	Agentic RAG, multi-agent orchestration, DRL pre-classification	Rule-based signatures + Welford $\mathcal{O}(1)$ + LightGBM T1, LangGraph Agent T2
Operational Goal	Centralized log aggregation and playbook response	Automated incident investigation via language reasoning	Resolve volume at cheap tier, reserve costly tier for complex cases
Latency & Cost Tiering	Single cheap tier, no context reasoning	Present in select works (DRL pre-classification); offload rate not reported against the attack base rate	Two-tier: 0.182 ms mean at T1; 97.5% offloaded before T2 at a 9.8% attack base rate
Data Sovereignty	On-premise local	Local deployment reported; hardware constraints not strictly bounded	100% Air-gapped on local GPU
Prompt Defense	Not applicable	Probabilistic classifiers (LanG: regex + Prompt Guard 2, 98.1% F1)	Structural Delimited Data Encapsulation, independent of payload wording
Forensic Audit	Standard text logs (susceptible to tampering)	Audit logs recorded (LanG: <code>mcp_audit.db</code>); no cryptographic sealing	HMAC-SHA256 tamper-evident evidence chain

Chapter 3

System Design and Technical Implementation

This chapter details the research methodology, followed by the system design and experimental implementation of SENTINEL: the two-tier processing pipeline, Tier 1 wire-speed filter and Machine Learning Gate, Tier 2 stateful LangGraph agent and Threat Memory, Dual-RAG hybrid knowledge retrieval, and adversarial AI security controls with cryptographic audit sealing.

3.1 Research Methodology

The thesis adopts the Design Science Research Methodology (DSRM) [36]. The core contribution of this research is a technical artefact, evaluated quantitatively through objective empirical metrics rather than purely conceptual arguments. The system is structured upon three core technical pillars:

1. *Rule-based, Statistical, and Machine Learning Pillar*: Provides wire-speed stream filtration via application signature rules (Rule-based), Welford $\mathcal{O}(1)$ online statistics [16], combined with a LightGBM Machine Learning Gate [17].
2. *Agentic Logic Pillar*: Provides deep contextual reasoning through a LangGraph state-graph [21] orchestrating a locally deployed Foundation-Sec-8B-Instruct model [19] over hybrid FAISS/BM25/RRF retrieval [23–25].
3. *Adversarial Security Pillar*: Establishes a robust defensive perimeter against log-substrate prompt injection risks [6, 8] and seals evidence records via HMAC-SHA256 signature chaining [29].

Operational efficacy of the SENTINEL architecture is quantitatively evaluated across five core measurement dimensions: classification quality, processing latency and throughput, adversarial resistance, explanation coherence (scored independently by a judge model), and forensic audit trail integrity. Comprehensive empirical results, comparative baselines, and system limitation verifications are detailed in Chapter 4.

3.2 SENTINEL Architecture Overview and SOC Stack Integration

SENTINEL operates under a cognitive two-tier architecture centered on computational cost-tiering. Raw telemetry from perimeter sensors streams into a Redis Stream queue and routes to Tier 1. Tier 1 serves as a stateless statistical and machine learning processing stage, resolving the majority of traffic volume at wire-speed. Among Tier 1’s action vocabulary, only ESCALATE proceeds to Tier 2, while BLOCK_IP, ALERT, AWAIT_HITL, and DROP terminate at Tier 1. This grounds the offload rate definition used in Chapter 4: an event is offloaded when resolved without triggering a language model invocation.

At Tier 2, the LangGraph agent [21] performs deep contextual reasoning over the Foundation-Sec-8B-Instruct model [19] (GGUF Q4_K_M quantized), integrating historical state from Threat Memory and evidence from Dual-RAG. Model outputs are bounded by a deterministic guardrail enforced to 3 action states (BLOCK_IP, LOG, AWAIT_HITL), with all records from both tiers sealed via HMAC-SHA256 signature chains [29]. SENTINEL operates as cognitive middleware between SIEM collectors and defense enforcement systems (NGFW/SOAR), absorbing stream noise at Tier 1 and providing cited justifications for escalated events at Tier 2, guaranteeing data sovereignty via air-gapped execution on a local GPU [7]. The complete processing flow and component interactions are illustrated in Figure 1.

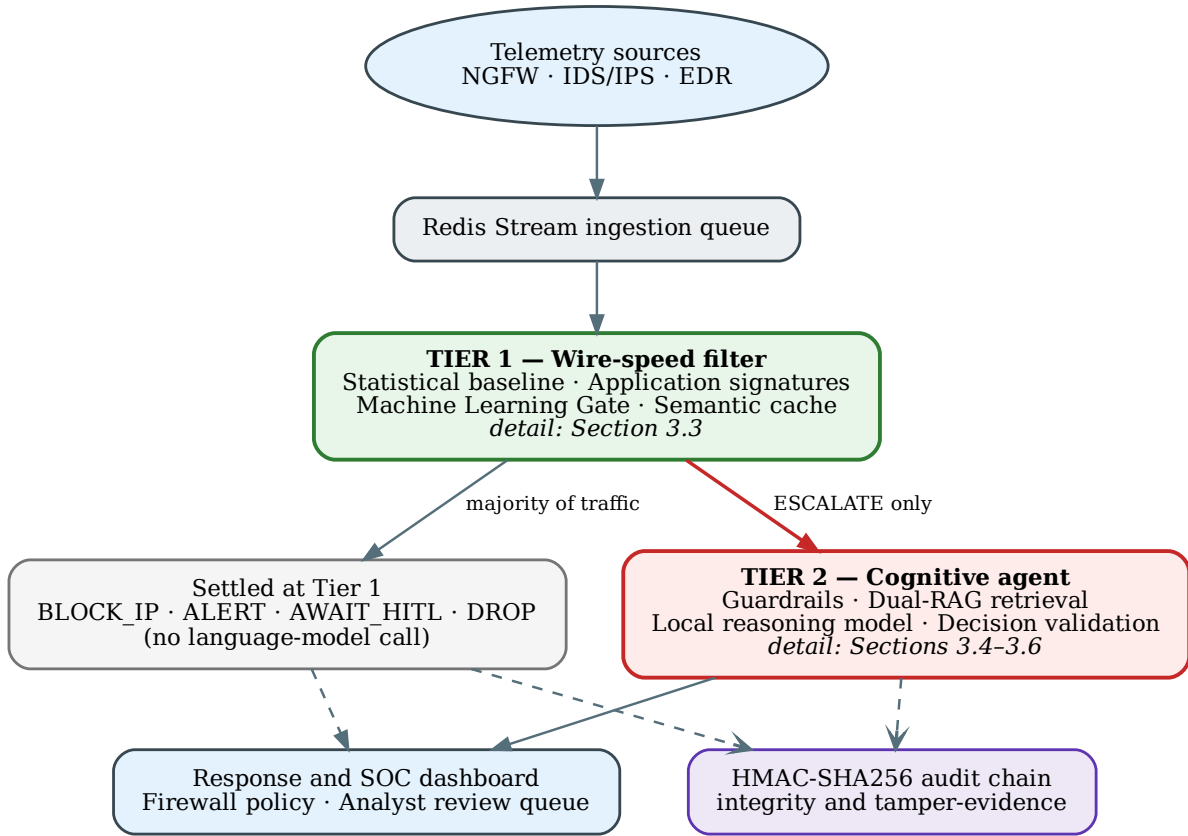


Figure 1. SENTINEL architecture overview.

3.3 Tier 1: Wire-speed Filter and Machine Learning Gate

Tier 1 operates as a stateless wire-speed perimeter layer combining application-layer signature rules and a Welford $\mathcal{O}(1)$ baseline statistical filter [16] (Equation 2.1). The signature engine covers 30 attack families normalized to OWASP CRS 3.3 benchmarks [37] (23 mapping to OWASP Top 10:2021; 7 targeting network/endpoint anomalies like ransomware and cryptomining). Concurrently, the Welford filter computes streaming Z-scores $Z = (x - \mu)/\sigma$ over flow attributes (packet rate, byte count, duration), updating baseline parameters μ and σ to discard conforming telemetry automatically. Detailed Tier 1 processing flows are shown in Figure 2.

For high-volume repetitive traffic (DDoS, credential stuffing), the Tier 1.75 Semantic

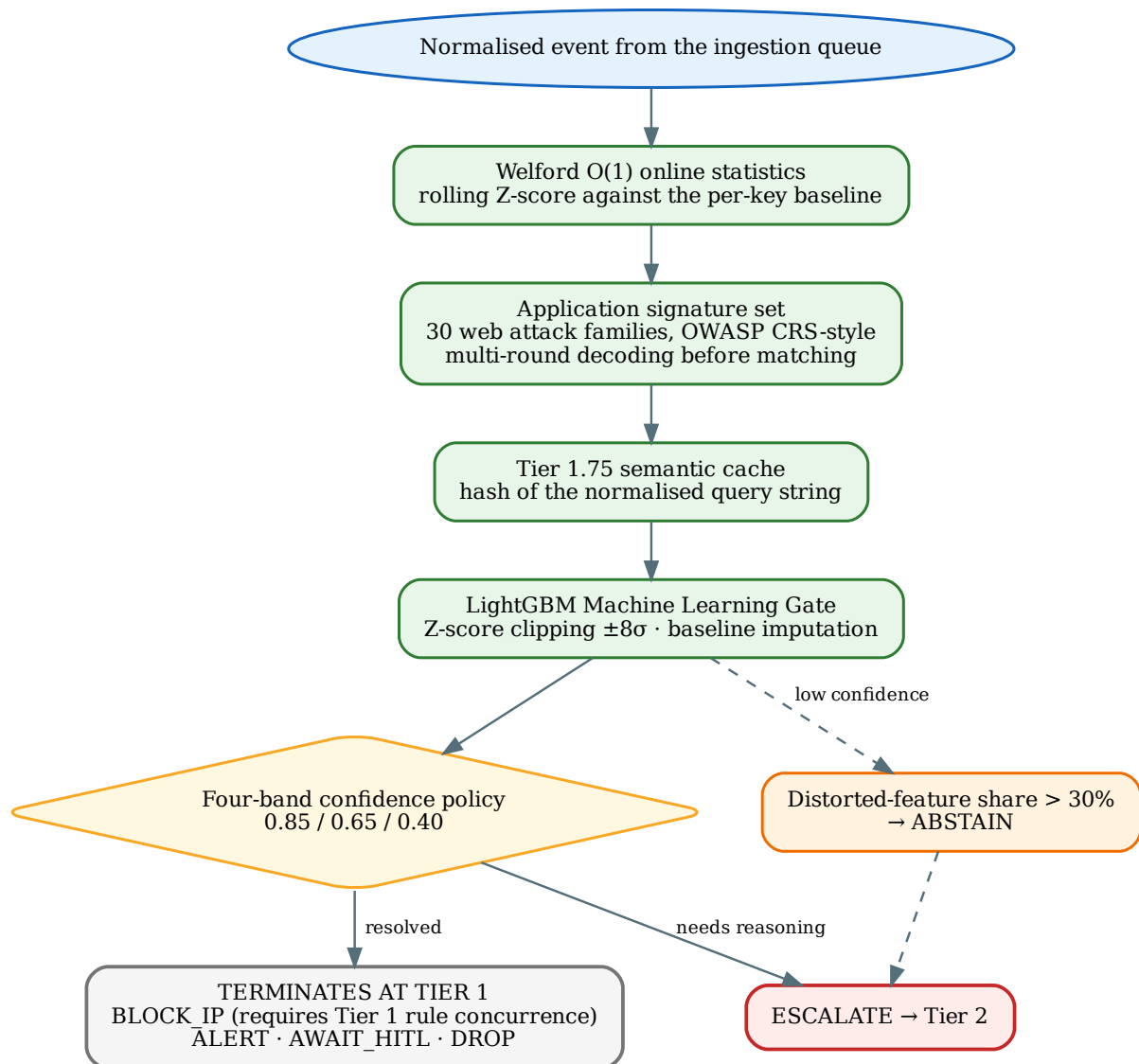


Figure 2. Tier 1 detail diagram.

Cache hashes normalized query strings in RAM, reusing prior decisions in < 1 ms without recalculation overhead.

Downstream, the LightGBM Machine Learning Gate [17] evaluates non-linear feature interactions, trained on 664,674 and tested on 189,907 NetFlow telemetry records drawn from the 949,535-record CSE-CIC-IDS2018 benchmark [9]. To counter evasion, the Gate clips Z-scores to $[-8, 8]\sigma$, imputes missing values with baseline means, and abstains to Tier 2 when distorted features exceed 30%. Adjudications map via a 4-band confidence policy (0.85 / 0.65 / 0.40 for BLOCK_IP / ESCALATE / ALERT / LOG), ensuring automated blocking executes only at highest confidence with signature concurrence.

3.4 Tier 2: LangGraph Cognitive Agent and Threat Memory

Tier 2 executes the Foundation-Sec-8B-Instruct model [19] (GGUF Q4_K_M) served by a CUDA-compiled llama.cpp engine on a local GPU. Reasoning flows are orchestrated by a LangGraph state-graph [21] across core stages: guardrail inspection (guardrails), context enrichment (rag_context), incident triage (llm_triage), technique mapping (attack_mapper), and action execution (action_executor) or human-in-the-loop hand-off (human_in_the_loop). The agent state-graph is illustrated in Figure 3.

At attack_mapper, absent explicit knowledge evidence, events route to analyst review (AWAIT_HITL) rather than emitting speculative actions. At llm_triage, ground-truth fields are stripped from the prompt for evaluation objectivity. Input context is managed via Drain log-template mining [31] paired with token budget monitoring.

To correlate multi-stage threat activities, Tier 2 maintains Threat Memory on SQLite (rollback-journal mode preventing Docker file locking). Reputation score S decays over silent period t : $S_{\text{new}} = S_{\text{old}} \times (1 - \lambda)^t$ where λ is the decay rate. Upon reaching $S = 100$, Tier 1 enforces permanent block-on-sight without re-triggering Tier 2. Critical internal infrastructure addresses are whitelisted to eliminate self-inflicted Denial of Service (Self-DoS) risks.

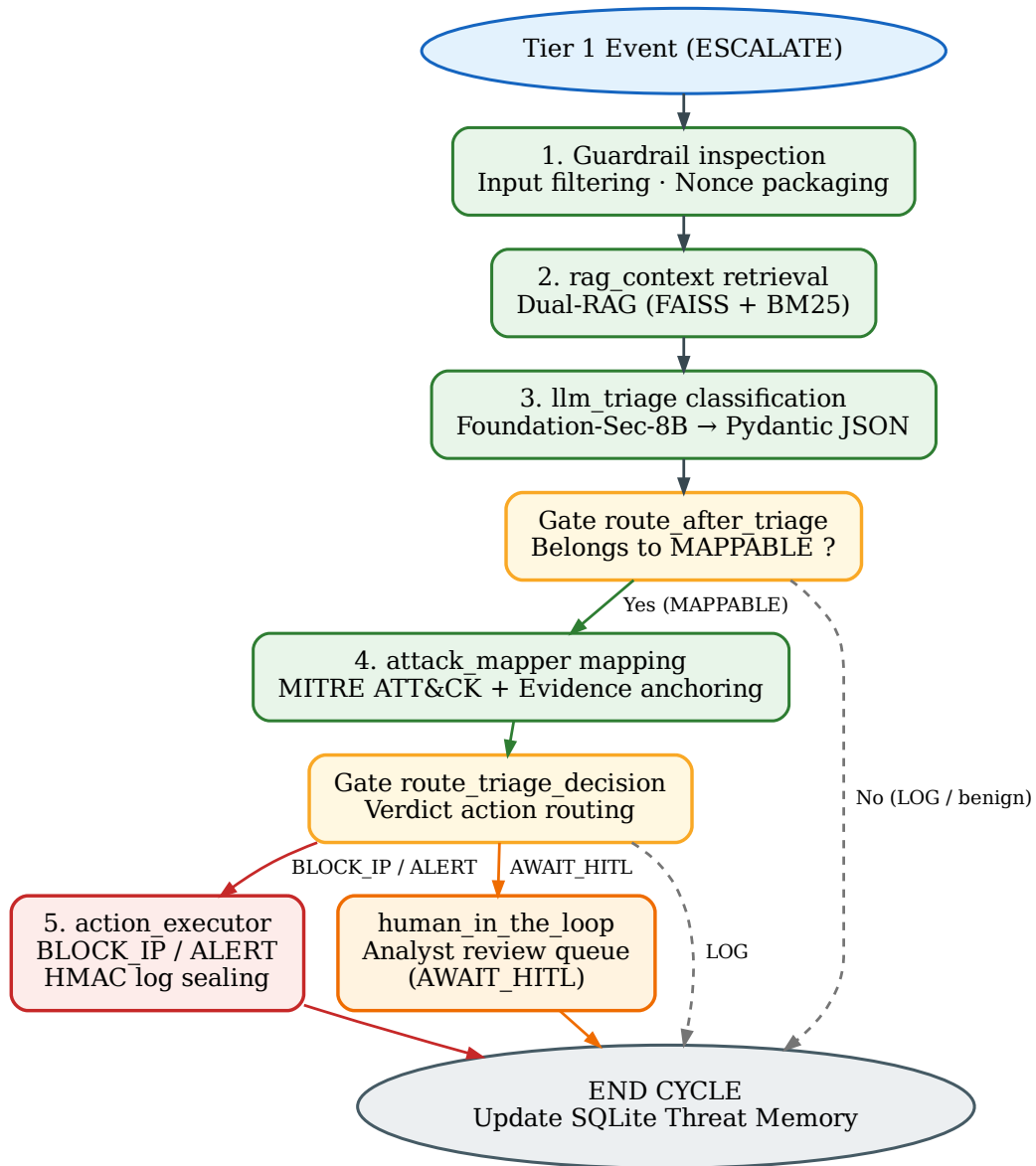


Figure 3. Tier 2 agent state-graph.

3.5 Dual-RAG Hybrid Knowledge Retrieval

To eliminate hallucinations, Tier 2 retrieves over a knowledge base of 433 STIX 2.1 MITRE ATT&CK entries [26] and NIST SP 800-61r2 guidelines [27]. Parallel channels combine FAISS [23] (Sentence-BERT [38]) for dense vector intent search with BM25 [24] for sparse keyword matching of CVEs and service ports. Results are merged via Reciprocal Rank Fusion (RRF, $k = 60$) [25] and injected at rag_context (Figure 4).

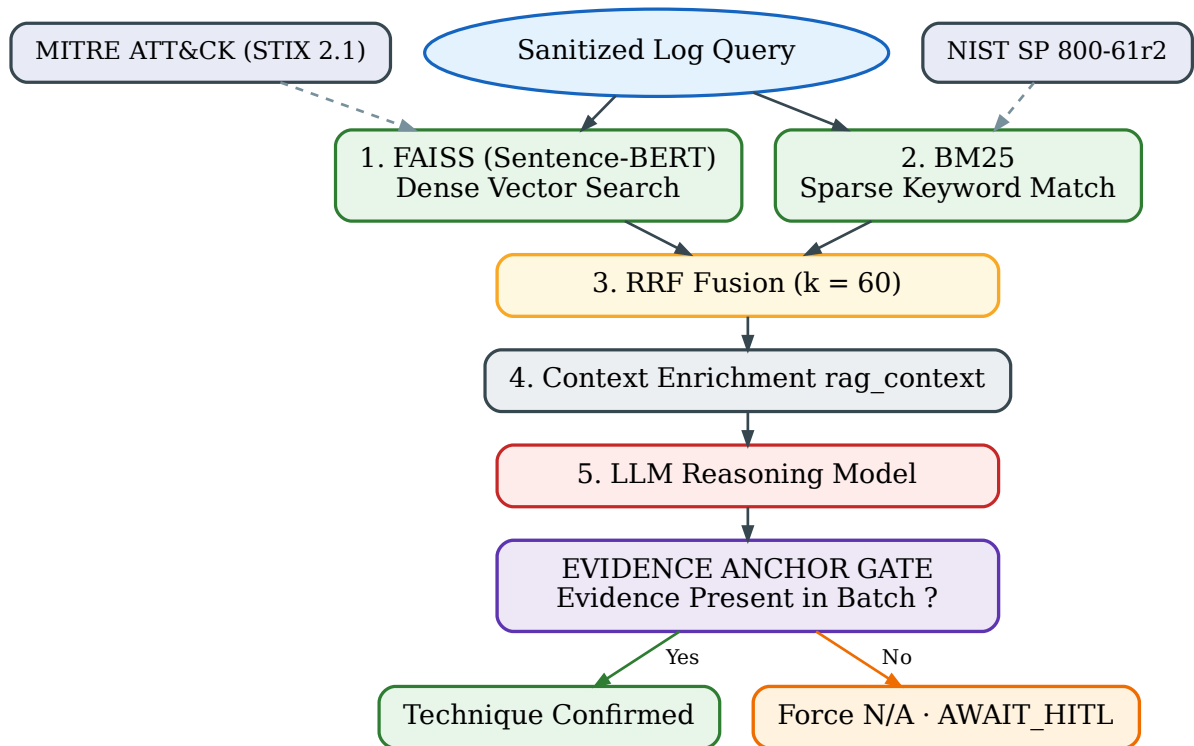


Figure 4. Dual-RAG hybrid retrieval and evidence-anchoring guardrail.

Retrieved text is sanitized by Unicode normalization, markup stripping, and command neutralization to prevent indirect injection. At output, an evidence-anchoring guardrail cross-references proposed MITRE ATT&CK codes against batch evidence. Unanchored codes are forced to N/A and deferred to AWAIT_HITL, so that no identifier absent from the retrieved evidence is ever acted upon.

3.6 Adversarial AI Security and Cryptographic Audit Sealing

Log prompt injection attacks [6, 8] embed control instructions inside raw log fields (User-Agent, URI). SENTINEL mitigates this via Delimited Data Encapsulation using dynamic per-batch random Nonce tokens, preventing delimiter escape (Figure 5). Three deterministic boundary validators guard system integrity: Data (filters malformed logs), Decision (enforces Pydantic LLMDecision and blocks internal IP bans), and Feedback (Zero Trust for AI rules).

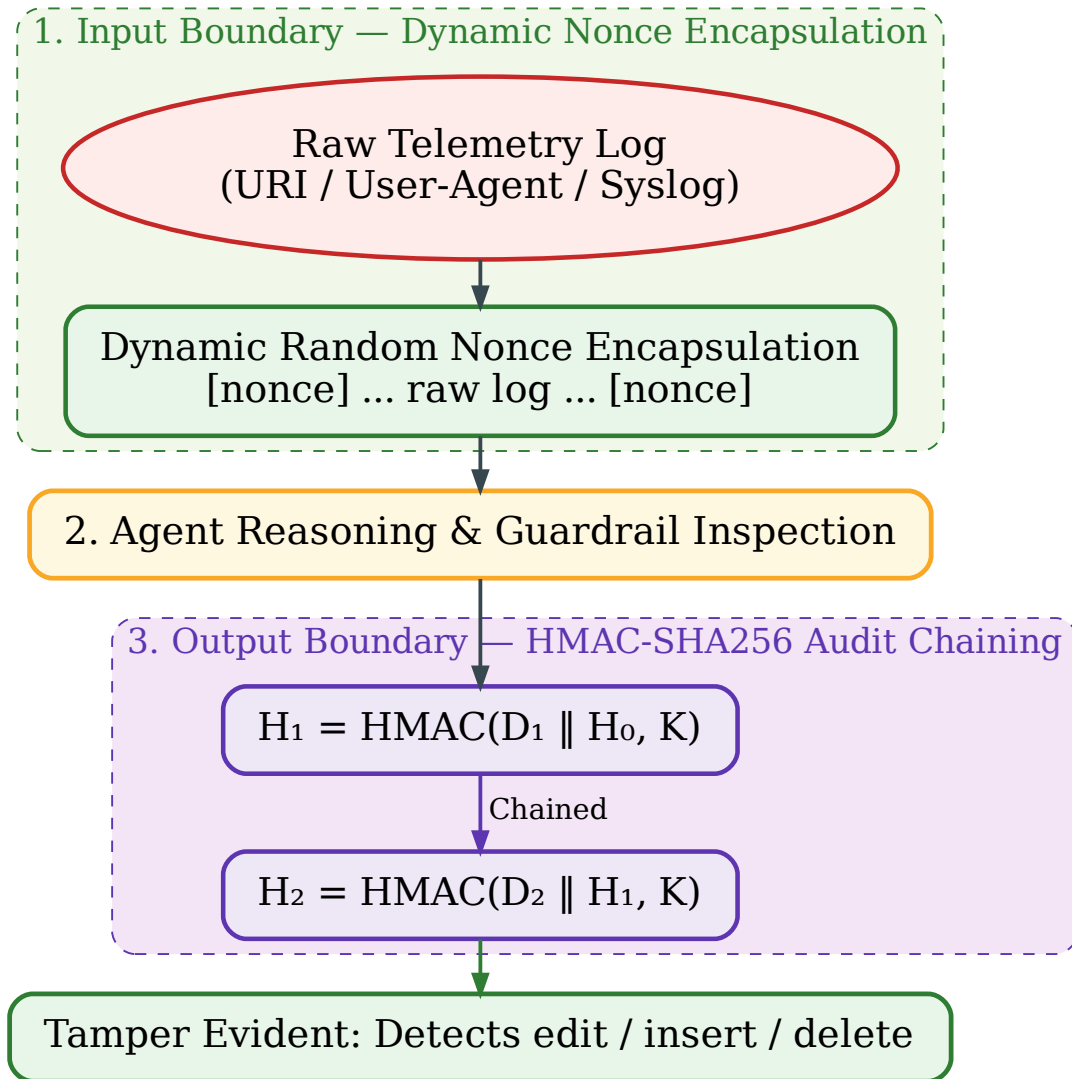


Figure 5. Input Nonce encapsulation and HMAC audit chaining.

For forensic auditing, each adjudication record D_i is cryptographically chained using HMAC-SHA256 [29]:

$$H_i = \text{HMAC-SHA256}(D_i \parallel H_{i-1}, K) \quad (3.1)$$

where H_{i-1} is the preceding record hash and K is the local secret key. Modifying, inserting, or deleting records breaks chain continuity and is flagged instantly, ensuring tamper evidence.

Chapter 4

Experiments and Evaluation

This chapter evaluates SENTINEL across three core dimensions: stream offloading performance, adversarial AI security guardrails with cryptographic audit integrity, and stateful agentic reasoning quality. Experiments were executed on an air-gapped environment containerized via Docker Compose (Intel Core i7, 32 GB RAM, local NVIDIA GPU running Foundation-Sec-8B-Instruct [19]). Offloading performance is measured on an operational stream of 99,717 events whose 9.8% attack base rate is consistent with the low proportion of genuine threats reported for production SOC [3], while security guardrails and classification quality are assessed across benchmark datasets (CSE-CIC-IDS2018 [9], CSIC 2010 [10], DAPT 2020 [11]) and 823 adversarial samples [12–14, 39]. All reported metrics state their respective denominators and address the core research questions directly.

4.1 Stream Offloading Performance and Filtration Economics

The primary bottleneck in deploying Large Language Models (LLMs) within Security Operations Centers (SOCs) is the high inference cost and latency incurred when evaluating every telemetry event individually. To address this challenge, SENTINEL implements a filtration economics architecture guided by the core design principle: “*resolve cheap adjudications deterministically before escalating to expensive reasoning*”. As defined in Section 3.2, an event is successfully offloaded when it is processed at lower filtration tiers without invoking LLM inference. This metric aggregates all terminal adjudication actions including DROP, BLOCK_IP, ALERT, and AWAIT_HITL, as none consume LLM inference tokens.

Empirical evaluation on the operational SOC-like stream of 99,717 events at a 9.8% attack base rate demonstrates SENTINEL’s capacity for computational workload reduction. The system successfully offloads 97.5% of total volume (eliminating 97,199 LLM invocations), escalating only 2.5% (2,518 events) to Tier 2 for cognitive reasoning. Table 2 summarizes the technical mechanisms, offload shares, mean latencies, and resource savings achieved across each processing tier of the system.

Table 2. Detailed technical mechanisms and per-tier workload offloading breakdown of SENTINEL (99,717-event SOC stream)

Processing Tier	Technical Mechanism	Offload Share	Mean Latency	Resource Savings Effect
Tier 1 (Rules & Welford)	Static WAF rules + $O(1)$ Z-score stats	38.3% (38,207 ev)	0.182 ms	0 LLM tokens, 0 GPU VRAM
Machine Learning Gate	LightGBM classifier (MCC 0.667)	59.2% (58,992 ev)	0.287 ms	58,992 LLM calls saved; 0 FP in 962 auto-blocks
Semantic Cache (Tier 1.75)	Exact-match HTTP payload fingerprinting; hit rate on RAG queries, not a share of the stream	81.3% (1,220/1,500)	9.8 ms	8.9× speed-up on duplicate queries
Tier 2 (Cognitive LLM)	LangGraph + Dual-RAG MITRE ATT&CK	2.5% (2,518 ev)	13,438 ms	Concentrates 95.0% threats into 15.76% volume (84.24% analyst reduction)
Cumulative (System)	Two-Tier Filtration Architecture	97.5% (97,199 ev)	Median: 0.88 ms	97,199 LLM calls eliminated (20,531× speedup vs. pure LLM)

The tiered offloading mechanism operates through close coordination among technical components. Tier 1 utilizes static signature rules combined with an $\mathcal{O}(1)$ Welford statistical algorithm [16] to resolve 38.3% of the stream at a mean latency of 0.182 ms without GPU consumption; that share comprises 31,012 benign events and 7,195 high-volume attacks. The LightGBM Machine Learning Gate [17] acts as an intermediate filter, evaluating each 76-dimensional feature vector in 0.287 ms (3,452 events/second) and offloading an additional 59.2% of benign telemetry. Evaluated on `datatest.json` (4,236 events containing flow features and HTTP payloads), the Gate achieves an MCC of 0.667 and a balanced ac-

curacy of 0.833 over the 2,534 events it adjudicates directly, together with 100% precision (0 false positives) across 962 autonomous blocking commands (BLOCK_IP). For repeated events, the Tier 1.75 Semantic Cache resolves 81.3% of queries with an $8.9\times$ speed-up factor. Consequently, Tier 2 receives only 2.5% of complex events, concentrating 95.0% of genuine threats into 15.76% of workload and reducing analyst reading material by 84.24%.

End-to-end latency evaluation over 500 events sampled at a fixed stride shows mean latency falling from 17,187.9 ms to 5,286.7 ms (a reduction of 69.2%; Mann–Whitney U test, $p \approx 9 \times 10^{-57}$). Median latency drops from 17,174.7 ms to 0.88 ms (nearly a $19,500\times$ reduction) as typical benign events are resolved under 1 ms at Tier 1. The 95th percentile latency (P_{95}) is 25,829 ms (vs. 21,434 ms under single-tier) because escalated events undergo pre-filtering before LLM processing, as shown in the Empirical Cumulative Distribution Function (ECDF) in Figure 6.

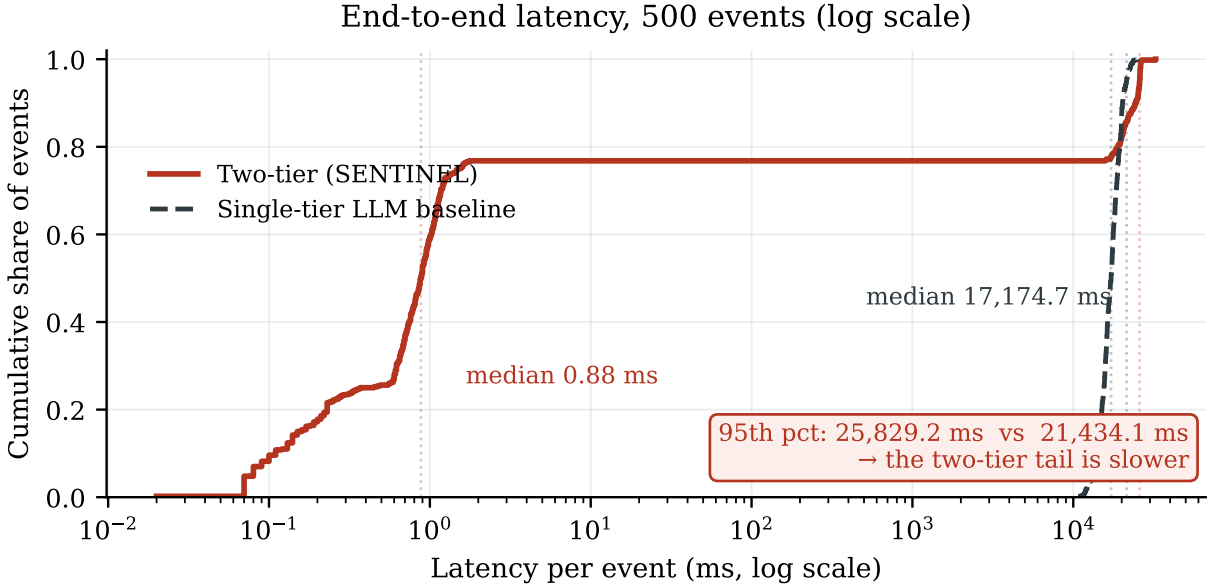


Figure 6. Cumulative latency distribution (ECDF) over 500 test events.

Throughput comparison on a 150-event stratified subset representing benign, network, and web attacks (Table 3) shows SENTINEL achieving 3,860 events/second, $20,531\times$ faster than a single-tier pure LLM baseline (0.19 events/second).

Table 3. Throughput comparison between SENTINEL and baseline configurations on 150 events

Control Configuration	Throughput (events/sec)
H1 — Static Signatures (30 WAF families)	11,129
SENTINEL — Tier 1 + Machine Learning Gate	3,860
H2 — Single-tier LightGBM (flat threshold)	2,162
H3 — Single-tier LLM (identical data, model, prompt)	0.19

4.2 Adversarial AI Guardrails and Cryptographic Audit Integrity

Adversarial security evaluation of SENTINEL is conducted across two primary threat surfaces: language-space prompt injection and numeric feature-space evasion. At Tier 1, the system deploys a primary static signature filtering layer comprising three parallel detectors (pattern matching, encoding neutralization, and delimiter isolation) to intercept malicious instructions early. This static layer is evaluated on 823 standardized prompt injection samples aggregated from three international benchmark corpora: AdvBench [12], deepset [13], and jackhhao [14], supplemented by DAN jailbreak variants [39]. Experimental results show that the static filter intercepts 18.8% (132/703 samples) of published benchmark corpora and achieves 72.5% (58/80 samples) on author-composed samples targeted at complex obfuscation techniques. When evaluated against a negative control set of 230 benign telemetry logs, the static layer records 46 false flags (a 20.0% rate); all 46 flags stem from the encoding detector and function strictly as enrichment metadata rather than causing denial-of-service (DoS) to legitimate traffic. Table 4 summarizes the evaluation results across all guardrail layers and cryptographic audit mechanisms.

For the remaining 678 complex adversarial samples escalating past Tier 1 (including 543 samples that genuinely bypassed static checks), the system routes telemetry to Tier 2. Rather than relying on probabilistic classifiers that inherently contain blind spots against novel attack patterns, Tier 2 enforces structural protection through Delimited Data Encapsulation. This mechanism encapsulates raw telemetry payloads within deterministic isolation boundaries using unique random nonces, preventing embedded instructions from overriding LLM system prompts. Evaluated over all 678 escalated samples (100% coverage),

Table 4. Adversarial security guardrail layers, evaluation denominators, and defence outcomes, including the cryptographic audit seals

Guardrail Layer & Threat Surface	Measurement Denominator	Defense Result
Tier 1 Static Filter (Published Corpora)	703 samples	18.8% (132/703)
Tier 1 Static Filter (Author-Composed Set)	80 samples	72.5% (58/80)
Tier 1 Static Filter (Benign Negative Control)	230 benign logs	20.0% false flags (annotation)
Machine Learning Gate (Extreme Feature Evasion)	1,036 attack samples	98.75% (1,023/1,036)
Machine Learning Gate (Abstention on Distorted Input)	4,236 events	116 T2 escalated events
Tier 2 Delimited Data Encapsulation (Injection)	678 escalated samples	100.0% (678/678)
Tier 2 Delimited Data Encapsulation (4 Positions)	100 published samples	100.0% (24/24, 23/23, 30/30, 23/23)
Pydantic Schema Validator (Induced Backend Error)	1 forced failure	AWAIT_HITL (no crash)
HMAC-SHA256 Seals (3 Tampering Scenarios)	30 trials per mode	100.0% tampering detection
HMAC-SHA256 Seals (Intact Log Control)	450 records	0 false alarms

Tier 2 achieves a 100% mitigation rate (678/678 samples), blocking every agent-hijacking attempt within this sample set. Position-specific tests across 100 samples distributed over four payload fields (URI, User-Agent, message, payload) confirm uniform 100% protection. When backend structural errors are deliberately induced, Pydantic schema validators safely drop the agent into an `AWAIT_HITL` state without system collapse.

On the numeric feature-space evasion surface, 1,036 attack samples derived from CSE-CIC-IDS2018 [9] and CSIC 2010 [10] datasets undergo extreme feature distortion to stress-test the LightGBM Machine Learning Gate. The Gate enforces Z-score clipping ($\pm 8\sigma$) and an autonomous abstention policy whenever over 30% of feature dimensions exceed 6σ . Consequently, 98.75% of attack samples (1,023/1,036) maintain correct malicious classification, while 116 anomalous events (from a 4,236 web/flow telemetry pool) are autonomously escalated to Tier 2 for deep cognitive adjudication. Figure 7 presents static pre-filter block rates by sample provenance alongside the false annotation rate on benign logs.

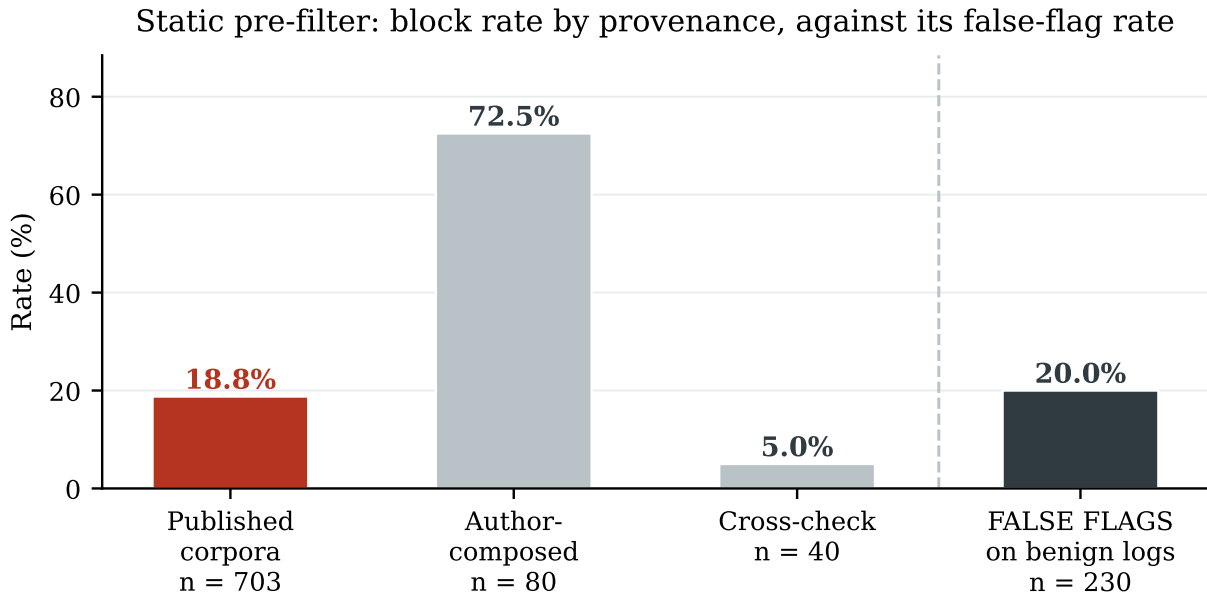


Figure 7. Static pre-filter block rate by sample provenance beside the false annotation rate on benign logs.

In parallel with adversarial guardrails, forensic audit log integrity is guaranteed via chained HMAC-SHA256 cryptographic signatures, where each log record incorporates the cryptographic hash of its predecessor. Empirical testing across 30 forgery attack trials for

each tampering category (content modification, fake record insertion, and intermediate record deletion) demonstrates 100% positional tampering detection accuracy with zero false alarms across 450 intact log entries, ensuring non-repudiation and forensic audit integrity for security operations.

4.3 Agentic Cognitive Reasoning and Attack Technique Attribution

SENTINEL’s cognitive reasoning capability and MITRE ATT&CK technique attribution performance are evaluated on a 500-payload sample subset extracted from CSIC 2010 [10] (comprising 250 attack-labeled web application payloads and 250 benign payloads). Experimental evaluation demonstrates that the hybrid Dual-RAG knowledge retrieval module (combining FAISS semantic search [23] and BM25 keyword search [24] via Reciprocal Rank Fusion [25]) achieves an 80.0% exact-match retrieval ceiling (200/250 samples) at the technique level, with a Hit@3 of 0.93 measured on the 243 of those samples that escalated. However, when evaluated across the end-to-end Tier 2 agent pipeline, technique attribution accuracy records 68.0% (170/250 samples, with tactic-level accuracy reaching 70.4% at a median latency of 13,293 ms). The 12.0 percentage-point drop from retrieval to end-to-end reasoning stems from the evidence-anchoring guardrail: 22 of the 30 samples lost were forced to N/A because the proposed technique was absent from the retrieved evidence documents. This represents an intentional architectural trade-off prioritizing ground-truth verification over ungrounded predictions. Table 5 details technique attribution accuracy and RAG retrieval ceilings.

Table 5. MITRE ATT&CK technique attribution accuracy and RAG retrieval ceiling (attribution on 250 labeled samples; retrieval on the 243 that escalated)

ATT&CK Technique	<i>n</i>	Hit@3	RAG Only	Full Pipeline
T1190 (Exploit Public-Facing Application)	52	1.00	100.0%	63.5%
T1059.007 (JavaScript Execution)	30	1.00	100.0%	73.3%
T1595.003 (Wordlist Scanning)	130	0.96	90.8%	86.9%
T1083 (File and Directory Discovery)	12	0.00	0.0%	16.7%
T1071.001 (Web Protocols)	26	1.00	0.0%	0.0%
Overall Technique Average	250	0.93	80.0%	68.0%

Table 5 demonstrates the empirical divergence between the RAG retrieval ceiling and end-to-end attribution accuracy. The 12.0 percentage-point gap (falling from the 80.0% retrieval ceiling to 68.0% end-to-end accuracy, corresponding to 30 misattributed events) decomposes into three distinct technical mechanisms:

First, the anti-hallucination role of the Evidence-Anchoring Guardrail: out of the 30 misattributed samples, 22 cases (accounting for 73.3% of the accuracy drop) occurred when the language model proposed an attack technique that was plausible under parametric intuition but absent from the five retrieved RAG documents injected into the context window. Rather than permitting ungrounded extrapolation, the guardrail strictly forced these decisions to N/A and routed them to the deferred queue (AWAIT_HITL). This represents a deliberate safety trade-off: forfeiting 8.8 percentage points of nominal autonomous accuracy to guarantee that 100% of emitted technique attributions possess verifiable evidentiary provenance, eliminating the operational hazard of automated firewall misconfigurations driven by AI hallucinations;

Second, hierarchical multi-label ambiguity inherent in web attack payloads: HTTP attack samples from CSIC 2010 [10] (such as SQL injection or cross-site scripting) simultaneously exhibit the characteristics of both public service exploitation (T1190) and client/system script execution (T1059.007). While the Dual-RAG module retrieves both reference documents, the ground-truth benchmark only records a single label. When the agent selects the accompanying valid technique, the single-label exact-match evaluation penalizes the decision as incorrect, bringing the full-pipeline accuracy of T1190 to 63.5% and T1059.007 to 73.3%;

Third, knowledge coverage disparity and context rank positions: for T1083 (File and Directory Discovery, $n = 12$), the retriever fails completely ($\text{Hit}@3 = 0.00$) due to knowledge base coverage gaps, denying the agent evidentiary grounding. Conversely, for T1071.001 (Web Protocols), the correct document is consistently retrieved at rank 3 ($\text{Hit}@3 = 1.00$), but reciprocal rank fusion (RRF) [25] weights fail to elevate it to rank 1, resulting in a 0.0% exact-match rate at both the retrieval and full-pipeline stages: a context-ranking disparity

rather than a knowledge coverage failure. Figure 8 compares the RAG retrieval ceiling against end-to-end attribution accuracy.

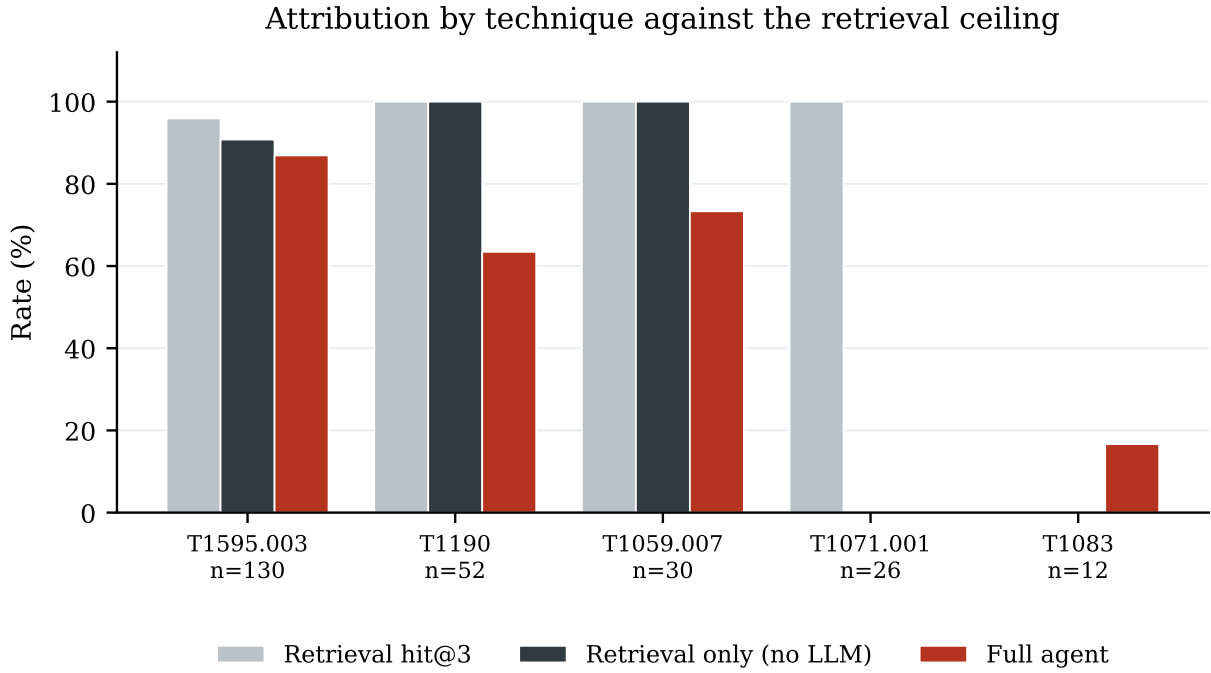


Figure 8. Per-technique attribution performance compared against RAG retrieval ceilings.

Tier 1 escalates 1,066 alerts to Tier 2, of which 80 represent true threats (7.5% baseline threat density). Through cognitive alert triage, the Tier 2 agent defers 168 alerts (15.76% of incoming volume) into a deferred queue (AWAIT_HITL). This queue encapsulates 76 out of 80 true threats (95.0% threat recall with a 45.24% alert precision, reflecting a $6.03\times$ enrichment factor over input). Consequently, SOC analysts reviewing only the deferred queue reduce reading volume by 84.24% while capturing 95.0% of true security threats. Figure 9 illustrates the Tier 2 alert triage funnel.

The reasoning quality of Foundation-Sec-8B-Instruct is evaluated on 1,052 escalated events along four RAGAS axes [40] by a judge model of a different family, Meta-Llama-3-8B-Instruct, following the recommended protocol against self-enhancement bias [41]. The evaluation yields an average score of 3.78/5.0 (answer relevance: 4.52, context recall: 4.11, faithfulness: 3.96, context precision: 2.54). Direct evidence anchoring, by contrast, reaches only 11.2% (95% CI 9.5–13.3%): fewer than one justification in eight cites a log value that can be checked back against the record, which bounds the transparency of the explanation

Tier 2 alert triage: the value is in the deferral channel, not the verdict

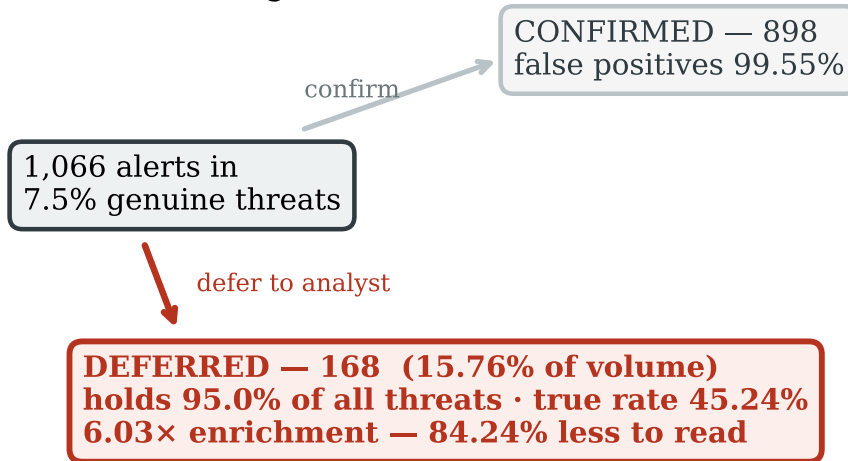


Figure 9. Tier 2 alert triage funnel comparing immediate decisions against deferred queues.

layer.

The relatively lower Context Precision score (2.54/5.0) compared to Context Recall (4.11/5.0) stems from three concrete architectural factors: First, semantic heterogeneity within the Dual-RAG knowledge corpus: the retrieval engine simultaneously pulls technical attack definitions from MITRE ATT&CK [26] and incident handling procedures from NIST SP 800-61r2 [27]. When the LLM judge evaluates pure attack attribution queries, all retrieved NIST passages (which detail containment, eradication, and recovery workflows) as well as adjacent sub-techniques are evaluated as non-target context, reducing the signal-to-noise ratio of the blended context window; Second, a deliberate recall-oriented retrieval policy: within operational SOC monitoring, missing an active threat (False Negative) due to an overly restrictive retrieval threshold poses an existential security hazard far greater than retrieving auxiliary reference text. The RRF hybrid search [25] intentionally sacrifices context purity to maintain high context recall (4.11/5.0, or 82.2%), delegating final discriminative filtering to the cognitive attention mechanism of the Tier 2 language model; Third, lexical overlap in standard network telemetry: cybersecurity event logs feature a high density of standardized protocol tokens (HTTP, POST, status code 200, standard headers). The BM25 keyword branch [24] within the hybrid fusion pipeline frequently pulls in techniques that share transport-layer primitives but differ in strategic intent.

Ablation evaluation across 1,700 test samples demonstrates that the full two-tier architec-

ture achieves an action accuracy of 35.35% (95% CI 33.12–37.66), significantly outperforming the single-tier Tier 1 baseline at 28.29% (95% CI 26.20–30.48). While single-tier Tier 1 reaches 69.01% self-resolution accuracy, it leaves 59.00% of events unaddressed; in contrast, the two-tier system leaves no event unresolved, at the cost of autonomous precision: 57.72% against 69.01%, with 55.24% autonomous actions and 44.76% explicit deferrals. Figure 10 compares action accuracy against the four RAGAS axes.

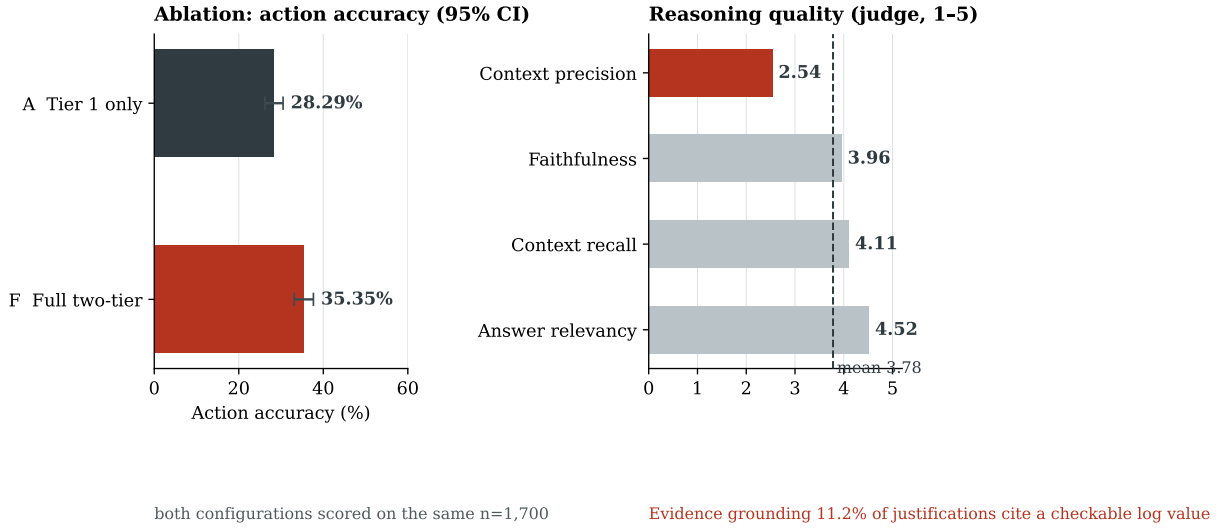


Figure 10. Component ablation action accuracy and the four RAGAS evaluation axes.

4.4 Infrastructure Scalability Analysis

Although the empirical measurements reported throughout this thesis were gathered on a single-GPU workstation, SENTINEL’s two-tier architecture provides a concrete architectural foundation to scale horizontally to enterprise production environments processing billions of events per day. To address how this prototype scales to production workloads, this section delineates between the empirical baselines established in this thesis and the literature-grounded architectural projections required for enterprise throughput.

Quantitatively, an enterprise telemetry workload of 1 billion events/day corresponds to a continuous mean throughput of $\approx 11,574$ events/second, with peak surges reaching 30,000 to 50,000 events/second during volumetric attack campaigns (such as distributed denial-of-service or distributed scanning). System scalability is substantiated across three archi-

tectural layers:

First, the stateless execution and linear compute complexity of Tier 1: application-layer static filtering, the $\mathcal{O}(1)$ Welford statistical model [16], and the LightGBM Machine Learning Gate [17] operate entirely on CPU cores with zero GPU VRAM consumption. With empirical throughput measured at 3,860 events/second on a single CPU core (Table 3), absorbing an enterprise baseline of 11,574 events/second requires only 3 to 4 distributed worker cores managed via standard message streaming partitions (e.g., Kafka Consumer Groups or Kubernetes Horizontal Pod Autoscaling). This projection is linear because Tier 1 maintains zero shared mutable state between events, eliminating lock contention. Furthermore, reimplementing Tier 1 in low-level systems programming languages (such as Rust or Go utilizing DPDK kernel-bypass frameworks to eliminate Linux socket buffer copying) provides an engineering projection exceeding 100,000 events/second on a single physical host, as documented in high-performance packet processing literature.

Second, the compounding offload efficiency of Filtration Economics and Drain3 clustering [31]: because Tier 1 empirically eliminates 97.5% of benign and commodity events (Table 2), escalated telemetry reaching Tier 2 is throttled to 2.5%, or approximately 25 million events/day (≈ 289 events/second). At Tier 2, the Drain3 log template miner [31] clusters recurring event formats (typical of repetitive scanning or automated probing) with compression ratios between $10\times$ and $50\times$, as repetitive payload patterns collapse into fixed parse trees. Consequently, the volume of unique events requiring deep cognitive inference by the language model is compressed down to a manageable 6 to 25 events/second.

Third, the architectural migration path for Tier 2 from workstation prototypes to enterprise clusters: the single-GPU sequential inference setup using `llama.cpp` serves as a minimal proof-of-concept environment (0.2 – 0.5 events/second with a median latency of 13,293 ms). In this single-threaded prototype (batch size $b = 1$), generation is strictly memory-bandwidth bound: the GPU spends the vast majority of execution time transferring the 8B model weights from VRAM to compute cores, resulting in low Tensor Core arithmetic utilization ($<15\%$). To satisfy an enterprise workload of 6 to 25 events/second,

Tier 2 is scaled through two principled software engineering deployments:

1. **Transition to High-Throughput Batching Inference Engines:** Migrating from sequential single-request execution to serving frameworks supporting PagedAttention and Continuous Batching, such as vLLM [42]. In the seminal benchmarks published by Kwon et al. [42] at ACM SOSP 2023, PagedAttention eliminates internal memory fragmentation (dropping waste from 60–80% down to under 4%) and dynamically packs tokens from concurrent request streams into single matrix multiplications. This shifts the inference bottleneck from memory bandwidth to compute cores, delivering an empirical throughput improvement of $15\times$ to $24\times$ over naive sequential serving for 7B–13B class models under realistic arrival traffic. Distributing this workload across a commodity multi-GPU node (e.g., 2 to 4 NVIDIA RTX 4090 or A10G GPUs with tensor parallelism) reliably provides the necessary aggregate capacity of 25 to 50 events/second. It is important to emphasize that this speedup represents an architectural projection derived from peer-reviewed literature benchmarks [42], rather than an in-situ empirical measurement on a physical enterprise cluster; physical validation under production multi-GPU load remains an ongoing engineering objective.
2. **Tiered Priority Queue Architecture:** Implementing an asynchronous message queue where critical-severity alerts (identified by extreme statistical anomaly scores from Tier 1) bypass standard buffers via a Fast Lane for immediate cognitive investigation, while ambiguous low-severity alerts are ingested via background batch processing. This ensures that critical incident response latency remains strictly bounded under burst traffic without overwhelming GPU memory.

In summary, the empirical evaluation reported above substantiates the practical utility of SENTINEL against the operational constraints of modern Security Operations Centers. Three architectural pillars carry that result—(1) high-speed filtration economics at Tier 1 that offloads 97.5% of raw telemetry, reduces median latency to 0.88 ms, and achieves a throughput of 3,860 events/second; (2) deterministic adversarial guardrails that resisted all 678 hard prompt-injection samples through Delimited Data Encapsulation, constrained

feature-space evasion, and sealed the forensic record with HMAC-SHA256 chaining [29]; and (3) evidence-anchored agentic cognitive reasoning at Tier 2 that isolates threats with 95.0% recall while reducing analyst workload by 84.24%—SENTINEL provides a reproducible basis for deploying agentic AI within enterprise security monitoring, subject to the limits reported above.

Chapter 5

Conclusion

This thesis has fundamentally resolved the critical bottleneck preventing the operational deployment of Agentic Artificial Intelligence (AI) within modern Security Operations Centers (SOCs): prohibitive inference costs, excessive processing latency, and technical hallucination risks under high-volume event streams. By designing, implementing, and empirically evaluating the SENTINEL system—a two-tier architecture operating completely locally on workstation GPU hardware—this research delivers three core scientific and practical contributions to Cybersecurity and AI Security.

First, the research establishes and validates the Filtration Economics paradigm in cybersecurity monitoring. By harmonizing application-layer static rules, an $\mathcal{O}(1)$ Welford statistical filter [16], and a LightGBM Machine Learning Gate [17], Tier 1 offloads 97.5% of raw SOC telemetry volume (eliminating 97,199 language model inference calls). This reduces median processing latency from 17,174.7 ms to 0.88 ms (a $19,500\times$ reduction in median latency) while achieving a throughput of 3,860 events/second with zero GPU VRAM consumption at Tier 1. Methodologically, the offload rate is reported against the base rate at which it was measured rather than as a constant of the system.

Second, the research institutes deterministic adversarial AI guardrails and cryptographically verifiable audit log integrity. Delimited Data Encapsulation at Tier 2 creates deterministic structural isolation boundaries, achieving 100% prompt injection mitigation across 678 complex adversarial samples in threat categories where static filters are entirely blind, thereby neutralizing agent hijacking risks. Concurrently, sequential HMAC-SHA256 cryptographic hash chaining [29] achieves 100% detection across the three core tampering modes (content modification, record insertion, and mid-chain deletion) with zero false

alarms on intact control chains, guaranteeing forensic audit integrity and non-repudiation.

Third, the research constructs a stateful agentic cognitive reasoning engine using LangGraph [21] integrated with Dual-RAG [22] and an Evidence-Anchoring Guardrail. The Tier 2 agent achieves an 80.0% MITRE ATT&CK knowledge retrieval ceiling (Hit@3 of 0.93) and 68.0% end-to-end technique attribution accuracy, with technique attribution constrained to identifiers present in the retrieved evidence. Simultaneously, through its alert triage funnel, the agent defers 15.76% of incoming alerts to isolate 95.0% of true threats (a $6.03\times$ threat density enrichment), enabling SOC analysts to reduce manual reading volume by 84.24%.

Alongside these empirical achievements, the SENTINEL architecture exhibits several technical limitations. End-to-end attribution accuracy (68.0%) remains constrained by RAG knowledge coverage gaps for rare attack techniques (such as T1083, which recorded a Hit@3 of 0.00), forcing evidence-anchoring guardrails to defer decisions to N/A and creating a 12.0 percentage-point gap relative to the retrieval ceiling in order to strictly eliminate hallucination hazards. Furthermore, RAGAS [40] context precision scores averaged 2.54/5.0 due to the intentional integration of dual NIST/MITRE knowledge streams and a deliberate recall-oriented retrieval policy (4.11/5.0 recall), while direct evidence anchoring to raw log lines reached 11.2% (95% CI 9.5–13.3%). Two architectural limitations further bound the present results: Tier 2 operates as a single LangGraph directed acyclic graph without role specialisation, so concurrent multi-stage campaigns are processed sequentially rather than in parallel; and every empirical measurement reported here was initially gathered on a single-GPU workstation, meaning that while Section 4.4 formalizes the quantitative scaling blueprint for enterprise workloads of billions of events per day, physical multi-GPU cluster validation remains an ongoing research objective.

These findings outline key avenues for future research. One primary direction is automating real-time RAG knowledge base ingestion from Cyber Threat Intelligence (CTI STIX/TAXII) feeds combined with context compression to eliminate Zero-day knowledge gaps. Another direction is domain-specific model fine-tuning (DPO) on telemetry log

datasets to enhance context mapping and enforce mandatory log line citations. A third direction addresses the single-graph limitation by extending Tier 2 into a multi-agent configuration with specialised roles for payload analysis, campaign correlation, and response execution. A fourth direction concerns throughput headroom at infrastructure scale: reimplementing the Tier 1 gate in a systems programming language and serving Tier 2 through vLLM PagedAttention [42]. Finally, integrating asymmetric digital signatures (Ed25519) and periodic external hash anchoring will elevate audit log integrity to full legal non-repudiation for critical cybersecurity infrastructures.

Concluding a rigorous research endeavor, while opening inspiring horizons in securing the digital realm through the power of Agentic Artificial Intelligence.

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